Lap Time Simulation: Methodology

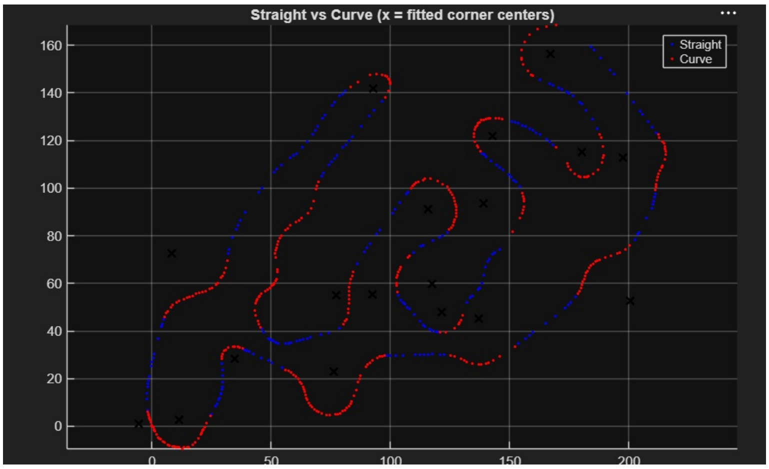
Overview

The lap time simulation splits a lap into two physically different problems: cornering, where speed is limited by available grip, and straights, where speed is limited by the powertrain and by how late the car can brake. Each is solved with a different model, then stitched together corner-to-straight-to-corner to build a full lap.

---

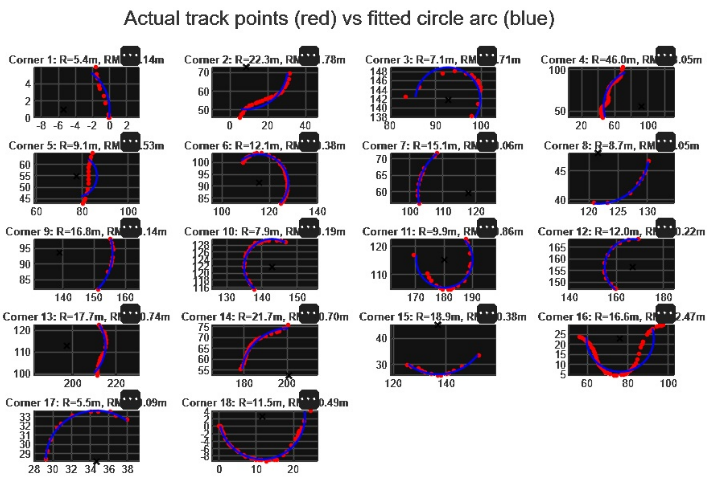
1. Track Reconstruction

Before any speed can be solved, the track itself has to be turned into geometry. Raw logged track points are classified as either straight or curved sections, then each curved section is fit to a circular arc to recover its radius and included angle.



Track segmentation

Corner centers (marked with an X) come out of this fit directly. This geometry, radius and angle per corner, is what feeds the cornering solve below.



Corner arc fits

Each panel compares the actual logged track points (red) against the fitted circular arc (blue), with fit radius and RMSE reported per corner. Some RMSE values are partially obscured by a UI overlay in the exported plot and weren't recoverable from this image; worth re-exporting cleanly if this goes into a final report.

---

2. Cornering Speed

For each corner, maximum speed is not assumed from a fixed friction circle; it's solved from a force balance that updates itself with speed. At a candidate speed, the model computes:

- **Aerodynamic downforce**, from the downforce coefficient-area and the candidate speed, split evenly across all four contact patches.
- **Lateral load transfer**, front and rear, computed separately using the corresponding axle's share of mass, the cornering acceleration at that candidate speed, CG height, and the front/rear track widths (front and rear are not assumed equal, since track width differs axle to axle).
- **Vertical load at each of the four wheels**, combining static load (from weight distribution), aero load, and load transfer, added on the loaded side and subtracted on the unloaded side.
- **Available lateral grip per wheel**, from a load-sensitive friction model where the friction coefficient itself falls as vertical load rises (grip doesn't scale linearly with load), then derated by a fixed factor to account for combined-slip and safety margin rather than assuming pure lateral capacity is fully usable.
- **Total available lateral force**, summed across all four wheels, compared against the lateral force actually required to hold that candidate speed through that corner's radius.

The candidate speed is adjusted by a root-finder until required and available lateral force match exactly, rather than being swept or guessed. Because downforce grows with the square of speed and load transfer also depends on speed, this equilibrium speed is genuinely coupled, not just a lookup, and has to be solved rather than computed in closed form.

Corner time then follows directly from arc length and solved speed.

---

3. Straight-Line Simulation

Straights are not solved with the same quasi-steady-state approach as corners; they're simulated as a full transient. Entry and exit speed for each straight are taken directly from the corner solve on either side, a corner's solved speed becomes the entry speed for the straight leaving it, and the exit speed for the straight entering it. The car then accelerates from its entry speed and must brake down to its exit speed before reaching the next corner.

The unknown is where along the straight the car switches from accelerating to braking. This transition point is found iteratively:

- A full vehicle acceleration model is run forward from the entry speed, giving a velocity-versus-time trace for the whole straight.
- Braking is not modelled as constant deceleration either. Like the acceleration phase, it's a full transient: speed is integrated backward from the required exit speed as a distance-domain ODE that includes aerodynamic drag as a speed-dependent term, so the deceleration rate itself changes as the car slows and drag falls off. This matters because a constant-deceleration assumption would

systematically under- or over-estimate the braking distance needed, especially at higher entry speeds where drag contributes meaningfully to slowing the car before the brakes take over.
- The transition point is adjusted (moved earlier or later along the straight) until the speed the accelerating phase reaches at that point matches the speed the braking phase requires at that same point, closing the gap between the two independently-simulated phases.

Total time on the straight is the sum of the accelerating-phase time and the braking-phase time at the converged transition point.

---

4. Total Lap Time

Cornering time is summed across every corner on the track, and straight-line time is summed across every straight, each carrying its own converged accelerate-then-brake transient. Total lap time is simply the sum of the two: the aggregate time spent grip-limited in corners, plus the aggregate time spent transitioning between the powertrain's acceleration limit and the brake system's (drag-inclusive) deceleration limit on the straights connecting them.